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April 8, 2026

Assignment 6

Assignment Overview:

The goal of this assignment is to research configuration management tools. This task involves researching different IT automation tools that can automate the application of STIGs, ensure the tools can also apply the DISA STIG. The final deliverable will be a report of your findings including screenshots of your results from the SCAP scan after applying the DISA STIGs via one of the automation tools listed. Further details are included below.

Main Tasks:

This project will encompass the following tasks:

1. Research at least 3 configuration management tools that can assist with automating OS hardening and include your findings in the report
2. Compare the configuration management tools, while consider the following:
 - **Agent vs Agentless Architecture**
 - i. **Chef and Puppet:** both use an agent based system which means that each machine being managed has a small program (called an agent) installed on it. This agent runs in the background and communicates with a central server to receive instructions and apply changes. The agent regularly checks in with the server, which makes it easier to manage systems continuously, but it also means extra setup and maintenance because you have to install and manage the agent on every machine.
 - ii. **Ansible:** use an agentless system which does not require any software to be installed on the target machines. Ansible connects to systems using SSH to run tasks. This makes it much easier to set up, especially for smaller environments or single machines like in this assignment. There is less overhead because there are no agents to install or update.
 - **Push vs Pull Models**
 - i. **Ansible:** uses a push model that the control machine sends commands directly to the target systems whenever you run a task. For example, when I run a playbook, Ansible immediately connects to the system and applies the changes. This is simple and gives me more control because I decide exactly when changes happen.
 - ii. **Chef and Puppet:** both use pull models the managed systems check in with a central server and pull their instructions from it. The agent on each system regularly contacts the server to see if there are any updates or

configurations it needs to apply. This is useful in large environments because systems stay up to date automatically without needing someone to manually run commands each time.

- **Declarative vs Procedural**

- Puppet:** use declarative approach that focuses on describing the end result you want. You basically tell the system what the final state should look like, and the tool figures out how to get there. Puppet is a good example of this because you define the desired state of the system, like making sure a service is running or a package is installed, and Puppet handles the steps automatically.
- Chef:** use procedural approach because it focuses more on the steps needed to reach that result. You write out the exact sequence of actions that should happen. Chef leans more toward this style because it uses a Ruby-based language where you can control how things are executed step by step.
- Ansible** is kind of in between. It uses playbooks that list tasks in order (which feels procedural), but many of its modules are designed to be idempotent, meaning they only make changes if needed (which is more like declarative behavior).

- **Scalability**

- Ansible** can scale pretty well, but it works best in small to medium environments. Since it uses a push model and connects over SSH, it can become slower when managing a large number of systems at the same time. However, it is still very useful because it is simple and doesn't require extra setup.
- Chef** is designed to scale better in large environments. Because it uses an agent and a central server, systems can automatically check in and apply updates without needing constant manual input. This makes it more efficient when managing many machines.
- Puppet** is also very scalable and is often used in big organizations. Its agent-based and pull model allows systems to continuously stay in the correct state, even as the number of machines increases. It is especially strong for long-term management and keeping systems consistent.

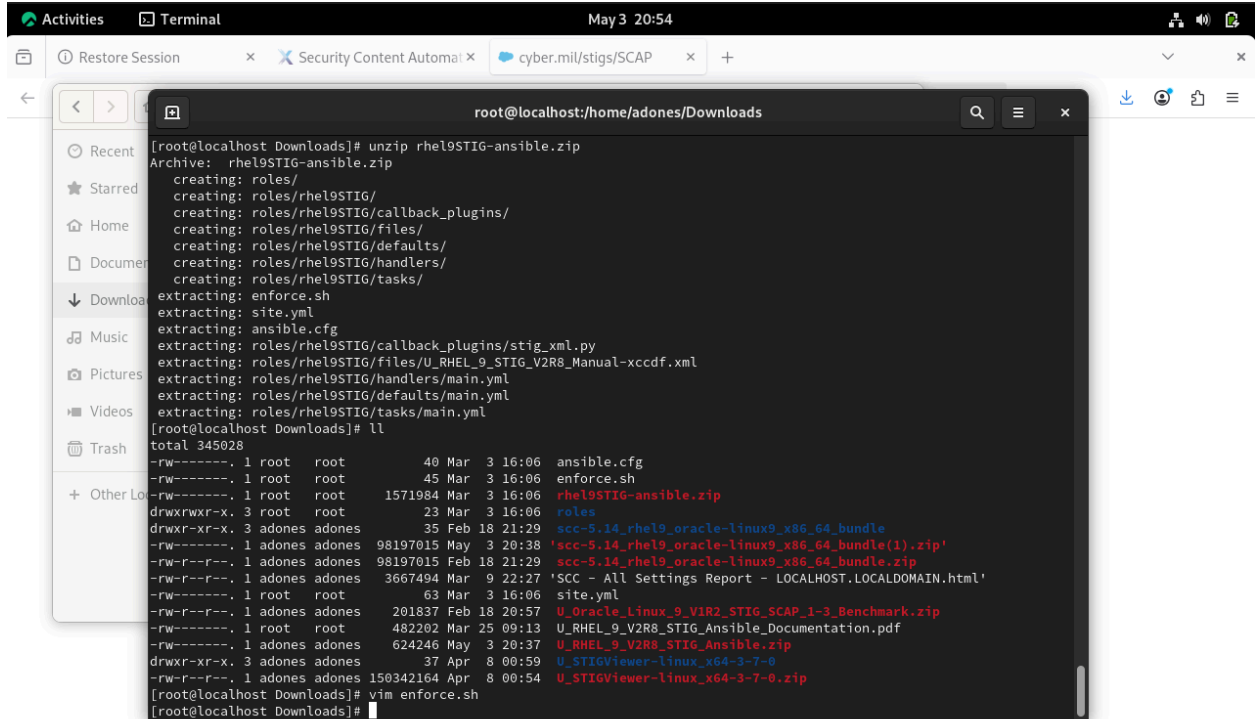
- **Ecosystem & Integration Capabilities**

- Ansible** has a very large ecosystem. It has Ansible Galaxy, where you can download ready-made roles and playbooks, including ones for security hardening like DISA STIGs. It also integrates well with cloud platforms, networking tools, and other automation systems. This makes it very flexible and easy to expand.

- ii. **Chef** also has a strong ecosystem, especially for enterprise environments. It includes tools like Chef Infra for automation and InSpec for compliance and security testing. Chef integrates well with DevOps pipelines and allows more advanced customization because it uses Ruby.
 - iii. **Puppet** has a mature ecosystem as well, with many modules available through the Puppet Forge. It is widely used in large organizations and integrates with enterprise tools for monitoring, reporting, and compliance. Puppet is especially strong when it comes to maintaining long-term system consistency.
3. Create a Fresh RHEL-Based OS virtual machine (RHEL/Rocky/Alma Linux, etc.). Either 8 or 9.
4. Install Ansible and apply the DISA STIGs using one of the playbooks.

```
[root@localhost adones]# dnf install ansible
Last metadata expiration check: 0:00:22 ago on Sun 03 May 2026 08:32:24 PM PDT.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
ansible                                noarch            1:7.7.0-1.el9     epel              34 M
Installing dependencies:
ansible-core                          x86_64            1:2.14.18-1.el9   appstream         2.2 M
git-core                             x86_64            2.47.3-1.el9_6    appstream         4.6 M
python3-cffi                          x86_64            1.14.5-5.el9      baseos            241 k
python3-cryptography                 x86_64            36.0.1-5.el9_6    baseos            1.2 M
python3-packaging                     noarch            20.9-5.el9        appstream         69 k
python3-ply                           noarch            3.11-14.el9.0.1    baseos            103 k
python3-pycparser                     noarch            2.20-6.el9        baseos            124 k
python3-pyparsing                     noarch            2.4.7-9.el9.0.1    baseos            150 k
python3-resolvelib                    noarch            0.5.4-5.el9       appstream         29 k
sshpas                                x86_64            1.09-4.el9        appstream         27 k
=====
Transaction Summary
=====
Install 11 Packages

Total download size: 43 M
Installed size: 405 M
Is this ok [y/N]: t
Is this ok [y/N]: y
Downloading Packages:
(1/11): python3-cffi-1.14.5-5.el9.x86_64.rpm                1.1 MB/s | 241 kB    00:00
(2/11): python3-cryptography-36.0.1-5.el9_6.x86_64.rpm     4.5 MB/s | 1.2 MB    00:00
(3/11): python3-ply-3.11-14.el9.0.1.noarch.rpm             1.5 MB/s | 103 kB    00:00
(4/11): python3-pycparser-2.20-6.el9.noarch.rpm            1.4 MB/s | 124 kB    00:00
(5/11): python3-pyparsing-2.4.7-9.el9.0.1.noarch.rpm       1.9 MB/s | 150 kB    00:00
(6/11): ansible-core-2.14.18-1.el9.x86_64.rpm             2.4 MB/s | 2.2 MB    00:00
(7/11): python3-packaging-20.9-5.el9.noarch.rpm            395 kB/s | 69 kB     00:00
(8/11): git-core-2.47.3-1.el9_6.x86_64.rpm                 4.1 MB/s | 4.6 MB    00:01
(9/11): python3-resolvelib-0.5.4-5.el9.noarch.rpm          513 kB/s | 29 kB     00:00
(10/11): sshpass-1.09-4.el9.x86_64.rpm                    422 kB/s | 27 kB     00:00
(11/11): ansible-7.7.0-1.el9.noarch.rpm                   13 MB/s | 34 MB     00:02
-----
Total                                                         12 MB/s | 43 MB     00:03
Extra Packages for Enterprise Linux 9 - x86_64                1.6 MB/s | 1.6 kB     00:00
Importing GPG key 0x3228467C:
Userid   : "Fedora (epel9) <epel@fedoraproject.org>"
Fingerprint: FF8A D134 4597 106E CE81 3B91 8A38 728F 3228 467C
From     : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-9
```



5. Install the SCC tool and run a scan

```

{"closed": "Documentation": "\man:chronyd(8)","\man:chrony.conf(5)","\DynamicUser": "no", "EnvironmentFiles": "/etc/sysconfig/chronyd (ignore_errors=yes)", "ExecMainCode": "0", "ExecMainExitTimestampMonotonic": "0", "ExecMainPID": "782", "ExecMainStartTimestamp": "Sun 2026-05-03 20:28:43 PDT", "ExecMainStartTimestampMonotonic": "11365338", "ExecMainStatus": "0", "ExecStart": "[ path=/usr/sbin/chronyd ; argv[0]=usr/sbin/chronyd $OPTIONS ; ignore_errors=no ; start_time=[n/a] ; stop_time=[n/a] ; pid=0 ; code=(null) ; status=0/0 ]", "ExecStartEx": "[ path=/usr/sbin/chronyd ; argv[0]=usr/sbin/chronyd $OPTIONS ; flags= ; start_time=[n/a] ; stop_time=[n/a] ; pid=0 ; code=(null) ; status=0/0 ]", "ExitType": "main", "FailureAction": "none", "FileDescriptorStoreMax": "0", "FinalKillSignal": "9", "FragmentPath": "/usr/lib/systemd/system/chronyd.service", "FreezerState": "running", "GID": "[not set]", "GuessMainPID": "yes", "IOAccounting": "no", "IOReadBytes": "18446744073709551615", "IOReadOperations": "18446744073709551615", "IOSchedulingClass": "2", "IOSchedulingPriority": "4", "IOWeight": "[not set]", "IOWriteBytes": "18446744073709551615", "IOWriteOperations": "18446744073709551615", "IPAccounting": "no", "IPEgressBytes": "[no data]", "IPEgressPackets": "[no data]", "IPIngressBytes": "[no data]", "IPIngressPackets": "[no data]", "ID": "chronyd.service", "IgnoreOnIsolate": "no", "IgnoreSIGPIPE": "yes", "InactiveEnterTimestampMonotonic": "0", "InactiveExitTimestamp": "Sun 2026-05-03 20:28:43 PDT", "InactiveExitTimestampMonotonic": "11091647", "InvocationID": "9457690768db49cd886b041ef7008fe3", "JobRunningTimeoutUSec": "infinity", "JobTimeoutAction": "none", "JobTimeoutUSec": "infinity", "KeyringMode": "private", "KillMode": "control-group", "KillSignal": "15", "LimitAS": "infinity", "LimitASSoft": "infinity", "LimitCORE": "infinity", "LimitCORESoft": "infinity", "LimitCPU": "infinity", "LimitCPUSoft": "infinity", "LimitDATA": "infinity", "LimitDATASoft": "infinity", "LimitFSIZE": "infinity", "LimitFSIZESoft": "infinity", "LimitLOCKS": "infinity", "LimitLOCKSSoft": "infinity", "LimitMEMLOCK": "8388608", "LimitMEMLOCKSoft": "8388608", "LimitMSGQUEUE": "819200", "LimitMSGQUEUESoft": "819200", "LimitNICE": "0", "LimitNICESoft": "0", "LimitNOFILE": "524288", "LimitNOFILESoft": "1024", "LimitPROC": "14359", "LimitPROCSoft": "14359", "LimitRSS": "infinity", "LimitRSSSoft": "infinity", "LimitRTPRIO": "0", "LimitRTPRIOSoft": "0", "LimitRTTIME": "infinity", "LimitRTTIMESoft": "infinity", "LimitSIGPENDING": "14359", "LimitSIGPENDINGSoft": "14359", "LimitSTACK": "infinity", "LimitSTACKSoft": "8388608", "LoadState": "loaded", "LockPersonality": "yes", "LogLevelMax": "-1", "LogRateLimitBurst": "0", "LogRateLimitIntervalUSec": "0", "LogsDirectoryMode": "0755", "MainPID": "782", "ManagedOOMMemoryPressure": "auto", "ManagedOOMMemoryPressureLimit": "0", "ManagedOOMPreference": "none", "ManagedOOMSwap": "auto", "MemoryAccounting": "yes", "MemoryAvailable": "infinity", "MemoryCurrent": "4374528", "MemoryDenyWriteExecute": "yes", "MemoryHigh": "infinity", "MemoryLimit": "infinity", "MemoryLow": "0", "MemoryMax": "infinity", "MemoryMin": "0", "MemoryPeak": "4919296", "MemorySwapMax": "infinity", "MountAPIVFS": "no", "NFileDescriptorStore": "0", "NRestarts": "0", "NUMAPolicy": "no", "Names": "chronyd.service", "NeedDaemonReload": "no", "Nice": "0", "NoNewPrivileges": "no", "NonBlocking": "no", "NotifyAccess": "none", "OOMPolicy": "stop", "OOMScoreAdjust": "0", "OnFailureJobMode": "replace", "OnSuccessJobMode": "fail", "PIDFile": "/run/chrony/chronyd.pid", "Perpetual": "no", "PrivateDevices": "no", "PrivateIPC": "no", "PrivateMounts": "no", "PrivateNetwork": "no", "PrivateTmp": "yes", "PrivateUsers": "no", "ProcSubset": "all", "ProtectClock": "no", "ProtectControlGroups": "yes", "ProtectHome": "yes", "ProtectHostname": "yes", "ProtectKernelLogs": "yes", "ProtectKernelModules": "yes", "ProtectKernelTunables": "yes", "ProtectProc": "invisible", "ProtectSystem": "full", "RefuseManualStart": "no", "RefuseManualStop": "no", "ReloadResult": "success", "ReloadUSec": "1", "RemainAfterExit": "no", "RemoveIPC": "no", "Requires": "sysinit.target -mount system.slice", "RequiresMountsFor": "/var/tmp", "Restart": "no", "RestartKillSignal": "15", "RestartUSec": "100ms", "RestrictAddressFamilies": "AF_INET AF_INET6 AF_NETLINK AF_UNIX", "RestrictNamespaces": "yes", "RestrictRealtime": "no", "RestrictSUIDSGID": "yes", "Result": "success", "RootDirectoryStartOnly": "no", "RuntimeDirectoryMode": "0755", "RuntimeDirectoryPreserve": "no", "RuntimeMaxUSec": "infinity", "RuntimeRandomizedExtraUSec": "0", "SameProcessGroup": "no", "SecureBits": "0", "SendSIGHUP": "no", "SendSIGKILL": "yes", "Slice": "system.slice", "StandardError": "inherit", "StandardInput": "null", "StandardOutput": "journal", "StartLimitAction": "none", "StartLimitBurst": "5", "StartLimitIntervalUSec": "10s", "StartupBlockIOWeight": "[not set]", "StartupCPUShares": "[not set]", "StartupCPUWeight": "[not set]", "StartupIOWeight": "[not set]", "StateChangeTimestamp": "Sun 2026-05-03 20:28:43 PDT", "StateChangeTimestampMonotonic": "11365368", "StateDirectoryMode": "0755", "StatusErrno": "0", "StopWhenUnneeded": "no", "SubState": "running", "SuccessAction": "none", "SyslogFacility": "3", "SyslogLevel": "6", "SyslogLevelPrefix": "yes", "SyslogPriority": "30", "SystemCallArchitectures": "native", "SystemCallErrorNumber": "2147483648", "SystemCallFilter": "~_sysctl afs_syscall bdflush break_chroot create_module delete_module finit_module fsconfig fsmount fsopen fspick ftime get_kernel_syms getpmg gty idle_init_module iopl_kexec_file_load_kexec_load lock_lookup_cookie modify_ldt mount mount_setattr move_mount mpx open_tree pciconfig_tlbasse pciconfig_read pciconfig_write perf_event_open pidfd_getfd pivot_root prof_profile ptrace putmsg query_module reboot rtas s390_pci_mmio_read s390_pci_mmio_write s390_runtime_instr security sgtestmask ssetmask stime stty subpage_ptr swapoff swapon switch_endian sysfs tuxcall ulimit umount2 uselib ustat vm86 vm86old vserver", "TTYReset": "no", "TTYVHangup": "no", "TTYVDIsallocate": "no", "TasksCurrent": "yes", "TasksMax": "22974", "TimeoutAbortUSec": "1min 30s", "TimeoutCleanUSec": "infinity", "TimeoutStartFailureMode": "terminate", "TimeoutStartUSec": "1min 30s", "TimeoutStopFailureMode": "terminate", "TimeoutStopUSec": "1min 30s", "TimerSlackNSec": "50000", "Transient": "no", "Type": "forking", "UID": "[not set]", "UMask": "0022", "UnitFilePreset": "enabled", "UnitFileState": "enabled", "UtmpMode": "init", "WantedBy": "multi-user.target", "WatchdogSignal": "6", "WatchdogTimestampMonotonic": "0", "WatchdogUSec": "0"}
TASK [rhel9STIG : stigrule_257946_etc_chrony_conf] *****

```

```

TASK [rhel9STIG : check if wireless network adapters are disabled] *****
ok: [localhost] => {"changed": false, "cmd": "[[ $(nmcli radio wifi) == 'enabled' ]]", "delta": "0:00:00.069147", "end": "2026-05-03 21:01:44.300925", "failed_when_result": false, "msg": "", "rc": 0, "start": "2026-05-03 21:01:44.231778", "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}

TASK [rhel9STIG : stigrule_258040_nmcli_radio_wifi_off] *****
changed: [localhost] => {"changed": true, "cmd": "nmcli radio wifi off", "delta": "0:00:00.077281", "end": "2026-05-03 21:01:44.910598", "msg": "", "rc": 0, "start": "2026-05-03 21:01:44.833317", "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}

TASK [rhel9STIG : stigrule_258041_etc_login_defs] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258043_etc_login_defs] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : run [[ $(sudo grep -i inactive /etc/default/useradd) != 'INACTIVE=35' ]] and register the result] ***
ok: [localhost] => {"changed": false, "cmd": "[[ $(sudo grep -i inactive /etc/default/useradd) != 'INACTIVE=35' ]]", "delta": "0:00:00.065742", "end": "2026-05-03 21:01:46.331093", "failed_when_result": false, "msg": "", "rc": 0, "start": "2026-05-03 21:01:46.265351", "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}

TASK [rhel9STIG : stigrule_258049_sudo_useradd__d__f_35] *****
changed: [localhost] => {"changed": true, "cmd": "sudo useradd -D -f 35", "delta": "0:00:00.056570", "end": "2026-05-03 21:01:46.822701", "msg": "", "rc": 0, "start": "2026-05-03 21:01:46.766131", "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}

TASK [rhel9STIG : stigrule_258054_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258055_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258056_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258057_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258060_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258069_etc_security_limits_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line added"}

TASK [rhel9STIG : stigrule_258070_etc_security_faillock_conf] *****
changed: [localhost] => {"backup": "", "changed": true, "msg": "line replaced"}

TASK [rhel9STIG : stigrule_258071_etc_login_defs] *****

```

```

root@localhost:/home/adones/Downloads

root@localhost:/home/adones/Downloads x root@localhost:/home/adones x
finity", "MemoryLimit": "infinity", "MemoryLow": "0", "MemoryMax": "infinity", "MemoryMin": "0", "MemoryPeak": "9662464", "MemorySwapMax": "infinity", "MountAPIVFS": "no", "NFileDescriptorStore": "0", "NRestarts": "0", "NUMAPolicy": "n/a", "Names": "NetworkManager.service", "NeedDaemonReload": "no", "Nice": "0", "NoNewPrivileges": "no", "NonBlocking": "no", "NotifyAccess": "none", "OOMPolicy": "stop", "OOMScoreAdjust": "0", "OnFailureJobMode": "replace", "OnSuccessJobMode": "fail", "Perpetual": "no", "PrivateDevices": "no", "PrivateIPC": "no", "PrivateMounts": "no", "PrivateNetwork": "no", "PrivateTmp": "no", "PrivateUsers": "no", "ProcSubset": "all", "ProtectClock": "no", "ProtectControlGroups": "no", "ProtectHome": "read-only", "ProtectHostname": "no", "ProtectKernelLogs": "no", "ProtectKernelModules": "no", "ProtectKernelTunables": "no", "ProtectProc": "default", "ProtectSystem": "yes", "RefuseManualStart": "no", "RefuseManualStop": "no", "ReloadResUlt": "success", "ReloadSignal": "1", "RemainAfterExit": "no", "RemoveIPC": "no", "RequiredBy": "NetworkManager-wait-online.service", "Requires": "dbus.socket system.slice sysinit.target", "Restart": "on-failure", "RestartKillSignal": "15", "RestartUsec": "100ms", "RestrictNamespaces": "no", "RestrictRealtime": "no", "RestrictSUIDSGID": "no", "Result": "success", "RootDirectoryStartOnly": "no", "RuntimeDirectoryMode": "0755", "RuntimeDirectoryPreserve": "no", "RuntimeMaxUsec": "infinity", "RuntimeRandomizedExtraUsec": "0", "SameProcessGroup": "no", "SecureBits": "0", "SendSIGHUP": "no", "SendSIGKILL": "yes", "Slice": "system.slice", "StandardError": "inherit", "StandardInput": "null", "StandardOutput": "journal", "StartLimitAction": "none", "StartLimitBurst": "5", "StartLimitIntervalUsec": "10s", "StartupBlockIOWeight": "[not set]", "StartupCPUShares": "[not set]", "StartupCPUWeight": "[not set]", "StartupIOWeight": "[not set]", "StateChangeTimestamp": "Sun 2026-05-03 21:04:47 PDT", "StateChangeTimestampMonotonic": "2174638746", "StateDirectoryMode": "0755", "StatusErrno": "0", "StopWhenUnneeded": "no", "SubState": "running", "SuccessAction": "none", "SyslogFacility": "3", "SyslogLevel": "6", "SyslogLevelPrefix": "yes", "SyslogPriority": "30", "SystemCallErrorNumber": "2147483646", "TTYReset": "no", "TTYVHangup": "no", "TTYVDisallocate": "no", "TasksAccounting": "yes", "TasksCurrent": "3", "TasksMax": "22974", "TimeoutAbortUsec": "1min 30s", "TimeoutCleanUsec": "infinity", "TimeoutStartFailureMode": "terminate", "TimeoutStartUsec": "10min", "TimeoutStopFailureMode": "terminate", "TimeoutStopUsec": "1min 30s", "TimerSlackNsec": "50000", "Transient": "no", "Type": "dbus", "UID": "[not set]", "UMask": "0022", "UnitFilePreset": "enabled", "UnitFileState": "enabled", "UtmpMode": "init", "WantedBy": "multi-user.target", "Wants": "network.target", "WatchdogSignal": "6", "WatchdogTimestampMonotonic": "0", "WatchdogUsec": "0"}}

RUNNING HANDLER [rhel9STIG : with_faillock_enable] *****
changed: [localhost] => {"changed": true, "cmd": "[authselect, \"enable-feature\", \"with-faillock\"]", "delta": "0:00:00.366025", "end": "2026-05-03 21:04:59.617798", "msg": "", "rc": 0, "start": "2026-05-03 21:04:59.251773", "stderr": "", "stderr_lines": [], "stdout": "Make sure that SSSD service is configured and enabled. See SSSD documentation for more information.", "stdout_lines": ["Make sure that SSSD service is configured and enabled. See SSSD documentation for more information."]}

RUNNING HANDLER [rhel9STIG : do_reboot] *****
fatal: [localhost]: FAILED => {"changed": false, "elapsed": 0, "msg": "Running reboot with local connection would reboot the control node.", "rebooted": false}

PLAY RECAP *****
localhost : ok=367 changed=240 unreachable=0 failed=1 skipped=2 rescued=0 ignored=0

[root@localhost Downloads]#

```

Through SCAP

Activities Sessions: 10 Files: 60 Total Size (MB): 75.59 May 3 21:21

SCAP Compliance Checker 5.14 [Configuration Profile: default - /opt/scc/options.xml]

File Options Results Help

Scan

1. Choose a scan type

Local Scan

2. Select Content

SCAP 1 of 4 Enabled

Show Content

3. Start Scan

Start Scan

View Results

Total Sessions 10

New Sessions 2

View Results

Output Error Warning

Manual Question Auto-Answer file for RHEL_9_STIG, version: 002.007.011 was not found in: /opt/scc/Resources/Content/Manual_Questions/Completed_Files/

All manual checks will be marked as 'Not Reviewed'.

* For CCCC Command Line Usage: Manual Question Auto-Answer templates for DISA STIG manual checks can be found in: /opt/scc/Resources/Content/Manual_Questions/Templates/

* For SCC GUI Usage: Manual Question Auto-Answer results for DISA STIG manual checks can be created by clicking on the 'Manual Questions' button in 'Content Details'

21:19:19 - Adjusted Score - 85.04% [YELLOW]
21:19:19 - Original Score - 85.04% [YELLOW]

Duration for scap content : RHEL_9_STIG on host LOCALHOST.LOCALDOMAIN 00:12:37

Analyzing Scan Session
Reading session data for 1 computers

Click 'View Results'

Creating Summary Viewer
Summary Viewer created at /home/adones/SCC/Sessions/2026-05-03_210640/SCC_Summary_Viewer_2026-05-03_210640.html

Total Errors: 0
Total Warnings: 1

Manual Question AutoAnswer file(s) were not found for: 1 SCAP Benchmarks, please review the scan session error log for more information

Review complete.
Session Results (XML, HTML, Text), if any, are located in the following directory:
/home/adones/SCC/Sessions/2026-05-03_210640/Results/

Session Logs (Debug, Screen, Error), if any, are located in the following directory:
/home/adones/SCC/Sessions/2026-05-03_210640/Logs/

Application Logs (Debug, Screen, Error), if any, are located in the following directory:
/home/adones/SCC/ApplicationLogs/

Computer Status Stream Status Current Stream

Log

09:06:42 PM: Attempting to validate original signature: U_RHEL_9_V2R7_STIG_SCAP_1-4_Benchmark-enhancedV11-signed.xml

09:06:42 PM: Digests matched for entire XML document.

09:06:42 PM: X.509 Certificate with Subject name "C=US, O=U.S. Government, OU=DoD, OU=PKI, OU=USN, CN=CS.NIWC-ATLANTIC.001" has been verified with a trusted public certificate

09:06:42 PM: Digital signature of file /opt/scc/Resources/Content/SCAP12_Content/U_RHEL_9_V2R7_STIG_SCAP_1-4_Benchmark-enhancedV11-signed.xml is valid

09:19:20 PM: Scan session completed : 2026-05-03_210640

09:19:23 PM: SCC Scan and Report Creation Complete

Activities SCC Report Viewer May 3 21:20

SCC Report Viewer [file:///home/adones/SCC/Sessions/2026-05-03_210640/Results/SCAP/LOCALHOST_SCC-5.14_2026-05-03_210640_All-Settings_RHEL_9_STIG-002.007.011.html]

Navigation Search

Back Forward Reload Search Match Case 0 of 0

All Settings Report - Red Hat Enterprise Linux 9 STIG SCAP Benchmark - NIWC Enhanced with Manual Questions

SCAP Compliance Checker - 5.14

Score | System Information | Content Information | Results | Detailed Results

Score

85.04%

Adjusted Score: 85.04%
Original Score: 85.04%
Compliance Status: YELLOW

Pass: 341	Not Applicable: 13	BLUE: Score equals 100
Fail: 60	Not Checked: 32	GREEN: Score is greater than or equal to 90
Error: 0	Not Selected: 0	YELLOW: Score is greater than or equal to 80
Unknown: 0	Informational: 0	RED: Score is greater than or equal to 0
Fixed: 0	Total: 446	

System Information

Target Hostname:	LOCALHOST.LOCALDOMAIN
Operating System:	RedHat Variant
OS Version:	9.7
Domain:	attlocal.net

6. Create a report with screenshots of your results from the scan performed. Include information about your experience using Ansible. Compare your past experience manually remediating to using the playbook.

In this assignment, I used my already hardened RHEL 9 virtual machine from my previous assignment and focused more on checking and improving its security. I first ran a scan to see the current compliance level, then used the Ansible STIG playbook to automatically fix any missing configurations. After that, I ran a second scan using the SCAP tool to compare the results and see if there were improvements. This helped me realize that even if a system is already hardened, there can still be some settings that need to be fixed. I also learned how much easier and faster it is to use automation instead of doing everything manually. Overall, this assignment helped me better understand how tools like Ansible and SCAP work together to improve system security and make sure it meets the required standards